

NAME:
NEPTUN:

First Solution: **B**
Second Solution: **D**

1. A stock price is currently \$800. Over each of the next two 6-month periods it is expected to go up by 25% or down by 20%. The risk-free interest rate is 10% per annum with continuous compounding. Suppose a trader sells 2000 1-year ATM European put options (20 contracts, each contract is for 100 stocks). What position in the stock is necessary to hedge the trader's position at the time of the trade?

- a) 707 (long) position in the underlying asset
- b) 707 (short) position in the underlying asset
- c) 1293 (long) position in the underlying asset
- d) 1293 (short) position in the underlying asset

$DF_{0,5}$ 0,9512
 q 56%

Stock
tree

$t = 0$	$t = 0,5$	$t = 1$
		1250
	1000	
800		800
	640	
		512

Put tree

$t = 0$	$t = 0,5$	$t = 1$
		0
	0,00	
50,82		0
	120,98	
		288

Long put delta
 $\Delta(0)$ -0,336

portfolio delta
 $\Delta(0)$ 672

Underlying asset delta
 $\Delta(0)$ -672

2. A stock price is currently \$800. Over each of the next two 6-month periods it is expected to go up by 25% or down by 20%. The risk-free interest rate is 10% per annum with continuous compounding. Suppose a trader buys 2000 1-year ATM European put options (20 contracts, each contract is for 100 stocks). A trader hedge the portfolio delta to zero in every time period. What position in the stock is necessary to hold if the the underlying asset price is decreasing during the first 6-month?

- a) 707 (long) position in the underlying asset
- b) 1293 (short) position in the underlying asset
- c) 0 position in the underlying asset
- d) 2000 (short) position in the underlying asset

$DF_{0,5}$ 0,9512
 q 56%

Stock tree			Put tree		
t = 0	t = 0,5	t = 1	t = 0	t = 0,5	t = 1
		1250			0
	1000			0,00	
800		800	50,82		0
	640			120,98	
		512			288

Long put delta		portfolio delta	
$\Delta_u(0,5)$	-1,000	$\Delta_u(0,5)$	2000

Underlying asset delta	
$\Delta_u(0,5)$	-2000